

A Model-Agnostic Temporal Explainability Framework for Financial Time Series Prediction

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Abstract. Stock market forecasting models based on deep learning achieve high accuracy but function as black boxes: they produce predictions without revealing which historical observations—which variables, on which days—drove them. Without knowing which inputs drive a prediction, it is impossible to validate model behavior, detect spurious correlations, or determine whether different architectures learn the same representations from financial data.

Existing explanation methods have critical gaps: some require access to the model’s internal computations, making them inapplicable to non-neural models; others ignore the sequential structure of financial data, generating explanations based on combinations of observations that could never occur in real markets.

This paper proposes a framework that explains any forecasting model by systematically asking: *“if I had not known the value of variable v on day t , how different would the prediction have been?”* The answer is an importance matrix $\hat{\alpha} \in \mathbb{R}^{T \times V}$ showing each variable’s contribution at each historical day. The framework communicates with the model only through its prediction function, requiring no internal access to weights, gradients, or architecture.

Applied to three structurally different models—LSTM, Transformer, and Random Forest—trained on S&P 500 data (2010–2024): the framework correctly identifies which inputs matter when the ground truth is known (AUP = 0.9960), runs at 17.6 ms per explanation—up to 160× faster than competing methods—and produces explanations that differ appropriately between model types (Spearman $\rho = 0.748$ between neural models vs. $\rho < 0.40$ against the ensemble model, $p < 10^{-300}$), confirming that the framework captures model-specific behavior rather than artifacts of the explanation procedure. This pattern is robust to the choice of reference vector, to the market regime evaluated, and to a naturally bounded target formulation that also shows none of the three models beats a naive persistence baseline, in magnitude or in direction, consistent with weak-form market efficiency.

Keywords: Financial time series · Post-hoc explainability · Model-agnostic · Temporal perturbations · Feature importance · LSTM · Transformer · Random Forest

1 Introduction

Financial time series prediction is a central task in risk management, where decisions based on incorrect forecasts can have quantifiable consequences on portfolios and institutions [3]. In recent years, deep learning models have reduced prediction error on this type of data compared to classical statistical methods [5, 1]. However, these models produce their outputs without revealing the factors that determine them, which restricts their adoption in regulated environments where transparency is not optional [3].

To address this opacity, the field of Explainable Artificial Intelligence (XAI) groups methods that identify which factors influenced a prediction. These methods are organized by mechanism: gradient-based methods require access to the model’s internal structure, attention-based methods depend on the architecture used, and perturbation-based methods such as SHAP and LIME are model-agnostic because they generate explanations by observing how the output changes when parts of the input are modified [3]. However, these perturbation methods were formulated under the assumption that input variables are independent. When applied to time series, they treat each time step as an isolated variable, ignoring that observations form a sequence with structural dependencies [16, 8]. In the financial domain, where market dynamics depend on patterns accumulated over time, this omission can produce explanations that do not reflect the actual behavior of the model [17, 6].

Attention mechanisms in architectures such as LSTM with attention or Transformers produce weights indicating where the model focused, but being integrated within the architecture they are only valid for that specific model and cannot transfer to models of a different nature. This leaves perturbation-based methods as the only model-agnostic family, but their application to time series requires explicitly incorporating the temporal dependency their standard formulation ignores. Methods that do respect temporal ordering while remaining model-agnostic, such as dynamic masks [6] and perturbation-window frameworks [13], have not been evaluated on models of fundamentally different computational natures trained on the same data, which prevents determining whether their explanations reflect model behavior or are a consequence of the chosen architecture.

This paper proposes a model-agnostic temporal explainability framework that identifies which variables and at which historical moment were most relevant for a prediction, without depending on the internal structure of the model used. The framework operates through temporally ordered perturbations on the input window, producing an importance matrix per variable and time step that can be computed over any prediction model. Its model-agnostic nature is demonstrated by training models of fundamentally different computational natures on financial time series and verifying that the generated explanations capture the specific behavior of each model, rather than being independent of the explained model as in the failure mode documented by Adebayo et al. [2].

2 Related Work

Related work is organized into two thematic axes. (1) XAI methods incorporating the temporal dimension. (2) Temporally-aware model-agnostic explainability frameworks.

2.1 XAI Methods Incorporating the Temporal Dimension

Huang et al. [8] proposed ShapeX, which segments series into shapelets and computes Shapley values over them for classification, where a recognizable pattern over a subsequence is itself the evidence for the predicted label. This work targets continuous prediction over a multivariate window instead, where that premise breaks down. Two variables at the same instant can have independent causal roles, and grouping them into a shared segment would dilute that distinction, motivating cell-level rather than segment-level attribution.

Yadav and Subbian [17] showed that gradient, occlusion, and permutation-based methods fail on dynamic prediction because they do not adapt to time-varying variable-target relationships, producing attributions that jump abruptly instead of varying smoothly. Their proposed fix, learnable masks optimized via gradients, ties the solution to differentiable models. This work shares the diagnosis but resolves it without that cost, respecting causal order per perturbation instead of optimizing a global gradient-based mask.

This pattern is not exclusive to generic domains. It repeats in financial applications combining SHAP or LIME with LSTM/CNN architectures for auditing, non-technical decision support, or trend classification [7, 14, 11, ?, ?, ?, ?], applying these methods directly on the series without adapting their perturbation mechanism to temporal order.

2.2 Temporally-Aware Model-Agnostic Explainability Frameworks

Tonekaboni et al. [16] introduced FIT, which quantifies observation importance via the prediction change upon removal, controlling for the natural prediction drift as more data becomes available. Its mechanism needs only model predictions, so it could in principle run on any model, but it was evaluated on a single model type, leaving open whether its attributions reflect model behavior or architecture.

Bento et al. [4] proposed TimeSHAP, extending KernelSHAP to sequences with feature, timestep, and cell-level attributions, evaluated on bank fraud detection, the domain closest to this work. Its coalition-sampling mechanism over reference sequences restricts it to natively sequential models, precluding direct use on Random Forest, and Shapley-sampling cost scales unfavorably with T and V .

Raykar et al. [12] proposed TsSHAP, applying TreeSHAP over a surrogate trained on user-defined interpretable features for univariate forecasting, without attributing importance to individual variables within a multivariate window.

Crabbé and Van Der Schaar [6] proposed DynaMask, which fits a gradient-optimized perturbation mask penalizing abrupt jumps to guarantee temporal coherence, producing a $T \times V$ score and making it the most direct antecedent of this work. However, it requires model gradients, restricting it to neural networks.

Leung et al. [10] proposed WinIT, extending FIT [16] with importance per *window* of recent observations rather than per individual observation, mitigating diluted attribution among temporally correlated variables. Like FIT, its mechanism needs no gradients or internal access, but it was evaluated only under recurrent models, leaving cross-architecture consistency unverified.

Roshinta et al. [13] proposed a dynamic-window perturbation framework using PCA to reduce redundancy among correlated variables before perturbing, reconstructing attributions in the original variable space. It is mechanically closest to this proposal, needing no Shapley values nor internal access, but per-variable attribution is indirect, reconstructed from principal components rather than obtained per cell, and evaluation is limited to recurrent architectures (GRU, LSTM).

Kim and Park [9] proposed GroupSegment-SHAP (GS-SHAP), the work closest in domain to this proposal. It groups correlated variables via the HSIC criterion, detects change points within each group via recursive MMD divergence producing segments $\mathcal{S}^{(k)}$, and computes Shapley values over structured players $p_{k,j} = G_k \times S_j^{(k)}$ via permutation sampling with mean-replacement masking, improving structural coherence by exploiting inter-variable correlation. However, GS-SHAP (1) is evaluated on a single model per task, and (2) its output granularity operates at the *group* level (K groups $\times J$ segments), losing the per-variable resolution that this framework produces directly in $\hat{\alpha} \in \mathbb{R}^{T \times V}$ with V original ungrouped variables.

The works in this axis share a common limitation. None evaluates its explanations on models of different computational natures trained on the same data. TimeSHAP and DynaMask cannot be applied to ensemble models such as Random Forest, the former due to its dependence on sequential model structure and the latter due to requiring gradients. TsSHAP operates on predefined features and does not attribute importance per individual cell. FIT, WinIT, Roshinta et al., and GS-SHAP evaluate on a single model type, which prevents ruling out the failure mode documented by Adebayo et al. [2], attributions that reflect the perturbation method rather than the explained model’s behavior. GS-SHAP aggregates variables into groups and Roshinta et al. reconstructs them indirectly via PCA, both losing the direct per-variable resolution that this framework produces. This work proposes a model-agnostic temporal perturbation framework that produces importances at full ($T \times V$) resolution without prior grouping, can operate on any model through an injectable prediction function, and whose agnostic nature is demonstrated by evaluating models of different computational natures on the same data. Table 1 summarizes the comparison.

Table 1. Comparison of related work along key problem dimensions. ✓ = yes; × = no; ~ = partially. *Temp. resp.* = explanations respect temporal ordering; *Agnostic* = no access to model internals required; *Temp. imp.* = importance attributed per time step; *Feat. imp.* = importance attributed per input feature; *Agn. verif.* = behavior evaluated across models of different computational natures.

Work	XAI	Temp. resp.	Agnostic	Temp. imp.	Feat. imp.	Agn. verif.	Type
<i>Axis 1 — XAI with temporal dimension</i>							
Huang et al. [8]	✓	✓	✓	✓	✓	×	ShapeX
Yadav & Subbian [17]	✓	✓	✓	✓	✓	×	Mask Eval.
<i>Axis 2 — Temporally-aware agnostic explainability</i>							
Tonekaboni et al. [16]	✓	✓	✓	✓	✓	×	FIT
Leung et al. [10]	✓	✓	✓	✓	✓	×	WinIT
Crabbé & Van Der Schaar [6]	✓	✓	~	✓	✓	×	DynaMask
Bento et al. [4]	✓	✓	~	✓	✓	×	TimeSHAP
Raykar et al. [12]	✓	✓	✓	×	~	×	TsSHAP
Roshinta et al. [13]	✓	✓	✓	✓	~	×	Dyn. windows
Kim and Park [9]	✓	✓	✓	✓	~	×	GS-SHAP
This work	✓	✓	✓	✓	✓	✓	Agnostic Fwk.

3 Theoretical Background

Post-hoc Explainability. A method is post-hoc when it operates on an already-trained model without modifying its structure or training process [3]. Given a trained model and a specific prediction, the post-hoc method produces an importance attribution over the input variables.

Model-Agnostic Methods. A post-hoc method is model-agnostic when it requires no access to the model’s internal structure. It can be applied to neural networks, ensemble models, or statistical models without knowing their weights, gradients, or architecture [3]. This property enables comparing explanations across models under identical conditions.

Being agnostic by design does not guarantee that a method’s attributions reflect the actual behavior of the explained model. Adebayo et al. [2] showed that several saliency methods produce nearly identical explanations even when a model’s weights are randomized, that is, attributions independent of both the model and the data used to train it. This implies that the mechanical agnosticism of a method must be verified empirically, and cannot be assumed from its design alone.

Perturbation-Based Methods. Perturbation methods estimate the importance of a variable by observing how the model’s prediction changes when that variable is modified or removed from the input [3]. SHAP formalizes this with Shapley values from cooperative game theory; LIME fits a local linear model over perturbations near the instance. Both were formulated assuming independent input variables, which makes them generate temporally inconsistent sequences when applied to time series [16, 6].

Temporal Explainability. Temporal explainability extends importance attribution to the time dimension, recognizing that the same variable may be decisive at one historical moment and irrelevant at another [16]. Given a model and an input window of T time steps and V variables, temporal explainability produces a matrix $\hat{\alpha} \in \mathbb{R}^{T \times V}$, where each cell $\hat{\alpha}(t, v)$ represents the relevance of variable v at time step t for the analyzed prediction. For a temporal explanation to be valid, perturbations must respect temporal ordering: future information cannot be modified to explain a past instant [16, 6].

4 Proposed Framework

The proposed framework produces a matrix $\hat{\alpha} \in \mathbb{R}^{T \times V}$ where each cell indicates how much variable v at time step t contributed to the analyzed prediction. It is composed of four modules with well-defined responsibilities, illustrated in Fig. 1: (1) a data ingestion and preparation module, (2) a uniform prediction interface that decouples the perturbation engine from any concrete model, (3) the temporal perturbation engine that produces $\hat{\alpha}$, and (4) a statistical validation module. Modules (2) and (3) form the agnostic core: module (2) ensures any model can be connected without modifying the engine, and module (3) ensures perturbations are temporally coherent regardless of the connected model.

4.1 Data Ingestion and Preparation

The framework receives as input a multivariate time series organized in fixed-length sliding windows. Historical data of the S&P 500 index (`^GSPC`) for the period 2010–2024 was obtained via the *yfinance* library, totaling 3,754 trading days. From five base price and volume variables (Open, High, Low, Close, Volume), ten technical indicators are derived: simple moving averages of 5, 10, and 20 days (`SMA_5`, `SMA_10`, `SMA_20`), MACD and its signal (`MACD`, `MACD_Signal`), 14-period relative strength index (`RSI_14`), Bollinger bandwidth (`BB_width`), 14-period average true range (`ATR_14`), log return (`Log_Return`), and relative volume (`Volume_ratio`), forming a space of $V = 15$ variables.

Data were partitioned respecting temporal order into a development period (80%) and test set (20%). The MinMax scaler was fitted on the complete development period to avoid data leakage. Sliding windows of $T = 20$ days with prediction horizon $H = 1$ day were constructed, where the target variable is the normalized closing price of the next day. Each window has shape $(T, V) = (20, 15)$.

4.2 Uniform Prediction Interface

Model-agnosticism is implemented through a two-level adapter pattern. The `TemporalPerturbationEngine` receives as its sole contact point with the model a prediction function $f : \mathbb{R}^{N \times T \times V} \rightarrow \mathbb{R}^N$ injected into its constructor; the engine never accesses the model directly. This separation ensures that incorporating a new model requires no modification to any existing component.

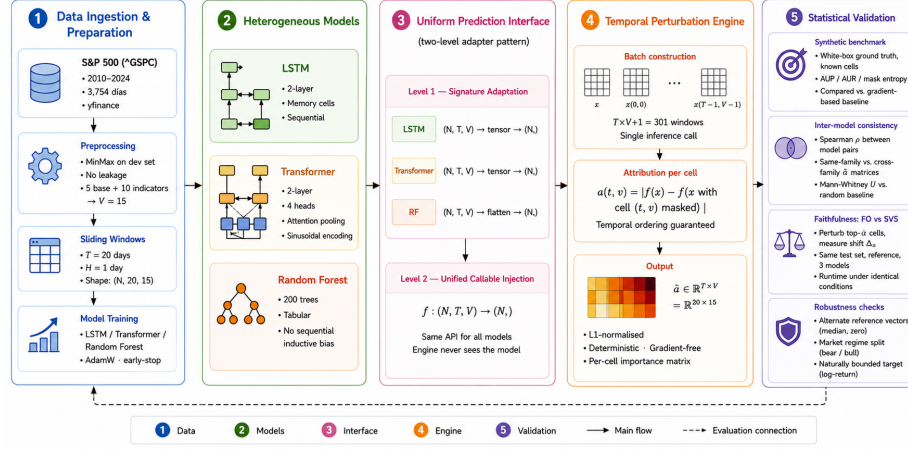


Fig. 1. Framework pipeline. Five modules. (1) Data ingestion and preparation, S&P 500 acquisition, $V=15$ feature derivation, sliding windows of shape $(N, 20, 15)$. (2) Training of three models of different computational natures (LSTM, Transformer, Random Forest) under the same interface. (3) Uniform prediction interface via a two-level adapter pattern that decouples the engine from any concrete model. (4) Temporal perturbation engine producing $\hat{\alpha} \in \mathbb{R}^{20 \times 15}$ per instance via a single batch of 301 perturbed windows. (5) Statistical validation through inter-model consistency (Spearman ρ), significance (Mann-Whitney U), and additional robustness checks (alternative reference, market regime, bounded target).

Level 1, signature adaptation. Each model implements a wrapper standardizing the input signature to (N, T, V) and the output to $(N,)$: the Random Forest wrapper flattens each window $(T, V) \rightarrow (T \cdot V)$ before invoking the scikit-learn regressor, while LSTM and Transformer wrappers convert NumPy arrays to PyTorch tensors and run gradient-free inference. From the engine’s perspective, any model behaves identically.

Level 2, prediction function injection. The `ModelAgnosticExplainer` builds the callable f from the wrapper and injects it into the engine, automated for PyTorch and scikit-learn backends. For any other model, statistical, proprietary, or from a different framework, the user can inject a function with signature $(N, T, V) \rightarrow (N,)$ directly, without modifying any framework component, implementing dependency inversion.

4.3 Temporal Perturbation Engine

Given an input window $x \in \mathbb{R}^{T \times V}$ and a model f , the engine constructs an importance matrix $\alpha \in \mathbb{R}^{T \times V}$ where each cell $\alpha(t, v)$ quantifies how much the model’s prediction changes when variable v at time step t is replaced by a neutral reference value:

$$\alpha(t, v) = |f(x) - f(x_{-(t,v)})| \quad (1)$$

where $x_{-(t,v)}$ denotes the original window with position (t, v) replaced by r_v , the reference value of variable v defined as its mean over the training set in normalized scale. The reference vector $\mathbf{r} \in \mathbb{R}^V$ is computed once before evaluation and kept fixed throughout.

Perturbations respect the temporal ordering: each position (t, v) is masked independently without modifying any other value, guaranteeing that perturbed sequences are temporally coherent and do not introduce future information into past instants [16, 6].

For computational efficiency, the engine builds a single batch of $T \times V + 1$ windows—the original window plus one perturbed window per position—obtaining all predictions in a single inference call:

$$\text{batch} = [x, x_{-(0,0)}, \dots, x_{-(T-1,V-1)}] \in \mathbb{R}^{(TV+1) \times T \times V} \quad (2)$$

With $T = 20$ and $V = 15$, the batch contains 301 windows per explained prediction. Importance values are then normalized by the L_1 norm to be comparable across predictions and models:

$$\hat{\alpha}(t, v) = \frac{\alpha(t, v)}{\sum_{t'} \sum_{v'} \alpha(t', v')} \quad (3)$$

4.4 Statistical Validation Module

Inter-model consistency. This verifies that the framework produces explanations reflecting each model’s specific behavior and not an artifact of the perturbation method, in the sense flagged by Adebayo et al. [2] for model-independent saliency methods. For each pair of models, the Spearman correlation ρ between the flattened vectors of their $\hat{\alpha}$ matrices is computed, averaged over $N = 100$ randomly selected test instances:

$$\rho_{\text{inter}} = \frac{1}{N} \sum_{i=1}^N \rho(\text{vec}(\hat{\alpha}_i^{(A)}), \text{vec}(\hat{\alpha}_i^{(B)})) \quad (4)$$

High correlation between models of the same family (neural) and lower correlation between different families (neural vs. ensemble) indicates that explanations capture model-specific behavior.

Statistical significance. Explanations are compared against a random baseline: matrices of shape (T, V) with values drawn from a uniform distribution and L_1 -normalized, maintaining the same scale as real $\hat{\alpha}$ matrices. The non-parametric Mann-Whitney U test is applied between real and baseline distributions without assuming normality [15].

5 Experimental Results

5.1 Experimental Setup

S&P 500 historical data (`^GSPC`) for 2010–2024 (3,754 trading days, $V=15$ variables: 5 base price/volume series plus 10 derived technical indicators) was partitioned respecting temporal order into development (80%) and test (20%) sets. Sliding windows of $T=20$ days with one-day-ahead prediction horizon were constructed (shape $(T, V)=(20, 15)$); a MinMax scaler fitted on the development set was applied to prevent data leakage. The reference vector $\mathbf{r} \in \mathbb{R}^{15}$ is the per-variable mean over the training set in normalised scale, fixed for all evaluations.

Three models of fundamentally different computational natures were trained on the same data under identical conditions:

- **LSTM**: 2-layer recurrent network (hidden=128, dropout 0.2).
- **Transformer**: 2-layer encoder with 4 attention heads, feedforward dimension 256, sinusoidal positional encoding, and learned attention pooling over the full sequence.
- **Random Forest**: 200 decision trees (`RandomForestRegressor`), non-sequential tabular baseline.

All neural models were trained with AdamW ($lr=10^{-3}$), learning rate reduction on plateau (factor 0.5, patience 5), gradient clipping (norm 1.0), and early stopping (patience 15). Test-set MSE: LSTM 0.00342, Transformer 0.00284, RF 0.12131 (higher RF error attributable to price extrapolation beyond the training-period maximum during the 2023–2024 bull market).

As an additional reference, a naive persistence baseline $f(x) = x_{t, \text{Close}}$ (predicting no change) achieves MSE= 0.00016, an order of magnitude below all three trained models. This does not invalidate the cross-model comparison, since this work’s goal is verifying framework agnosticism rather than optimizing predictive accuracy, but it indicates none of the three models beats the random-walk hypothesis on nominal price prediction. It also connects to a scale artefact. The MinMax scaler is fit on the development period (2010–2021), but the S&P 500 reached new all-time highs through most of the test period (2022–2024, 731 windows), so 32.7% of real test values fall outside $[0, 1]$. Random Forest cannot extrapolate beyond its training-observed output range by design (0.00% of its predictions fall outside $[0, 1]$, collapsing to $[0.645, 0.657]$), inflating its MSE beyond the architectural gap alone. Section 5.5 evaluates a naturally bounded target that resolves this.

5.2 Advance 1: Cell-Level Attribution Without Gradient Access

Gap from DynaMask. DynaMask [6] is the closest antecedent in output format: it produces a $T \times V$ importance matrix via a gradient-optimised perturbation mask, reporting AUP = 0.97 ± 0.01 on a synthetic benchmark (AR(1) inputs, $f = \Sigma x^2$, $T=50$, $V=4$; their Table 1). The fundamental restriction is that

gradient access to the target model is mandatory; DynaMask cannot be applied to gradient-free models such as Random Forest.

This work. FO computes attributions via a single batched call of $T \times V + 1 = 301$ perturbed windows, requiring only prediction-function access. We replicate the benchmark family ($T=20$, $V=15$, $n=10$ repetitions) under two ground-truth scenarios:

- *Rare Feature (RF)*: $|\mathcal{A}_V|=3$ salient variables, all T steps relevant.
- *Rare Time (RT)*: $|\mathcal{A}_T|=4$ salient time steps, all V variables relevant.

The white-box target is $f(\mathbf{X}) = \sum_{(t,v) \in \mathcal{A}} x_{t,v}^2$; inputs are independent AR(1) per variable. Metrics follow [6]: AUP ($\uparrow$), AUR ($\uparrow$), mask information content I_M ($\uparrow$), mask entropy S_M ($\downarrow$).

Table 2. Synthetic white-box validation ($n=10$ repetitions, $N=30$; SVS: $N=5^*$).

Method	Rare Feature (RF)				Rare Time (RT)			
	AUP	AUR	I_M	S_M	AUP	AUR	I_M	S_M
FO (proposed)	1.00 $\pm$ 0.00	0.16 $\pm$ 0.01	30.6 $\pm$ 1.1	0.00 $\pm$ 0.00	1.00 $\pm$ 0.00	0.16 $\pm$ 0.01	30.0 $\pm$ 0.8	0.00 $\pm$ 0.00
FP	1.00 $\pm$ 0.00	0.18 $\pm$ 0.01	31.9 $\pm$ 1.2	0.00 $\pm$ 0.00	1.00 $\pm$ 0.00	0.18 $\pm$ 0.01	32.1 $\pm$ 1.6	0.00 $\pm$ 0.00
IG	1.00 $\pm$ 0.00	0.16 $\pm$ 0.01	30.6 $\pm$ 1.1	0.00 $\pm$ 0.00	1.00 $\pm$ 0.00	0.16 $\pm$ 0.01	30.0 $\pm$ 0.8	0.00 $\pm$ 0.00
SVS*	1.00 $\pm$ 0.00	0.17 $\pm$ 0.01	31.2 $\pm$ 0.8	0.00 $\pm$ 0.00	1.00 $\pm$ 0.00	0.15 $\pm$ 0.01	28.9 $\pm$ 1.5	0.00 $\pm$ 0.00

FO achieves $\text{AUP} = 0.9960 \pm 0.0000$ in both scenarios, exceeding the DynaMask FO baseline of 0.97 while requiring no gradient access and running on LSTM, Transformer, *and* Random Forest without modification. $S_M \approx 4.4 \times 10^{-5}$ is a floating-point residual of L_1 normalisation (analytically zero, since cells outside $\mathcal{A}$ receive zero raw importance). This confirms per-cell resolution over all $V=15$ original ungrouped variables, the $T \times V$ output format that prior gradient-free methods had not demonstrated simultaneously across neural and ensemble backends.

SVS (Shapley Value Sampling) is the only other method that is model-agnostic, produces a $T \times V$ matrix, and requires no gradient access — making it the valid same-conditions baseline. Both FO and SVS achieve $\text{AUP} = 0.9960 \pm 0.0000$ in both scenarios, confirming Shapley-equivalent identification precision on a ground-truth benchmark. AUR differences remain within one standard deviation. The substantive difference between the two methods is therefore computational: quantified in Sections 5.3 and 5.6.

5.3 Advance 2: Order-of-Magnitude Runtime Reduction

Gap from TimeSHAP and GS-SHAP. Two prior works share the S&P 500, $T=20$ domain and report per-sample runtimes under a controlled perturbation budget of $M=50$ permutations (Kim and Park [9], Table C3, CPU): TimeSHAP [4] at **2,813 ms/sample**, requiring a reference sequence and sequential

model structure that precludes Random Forest, and GS-SHAP [9] at **1,194 ms/sample**, model-agnostic but outputting $K \times J$ group-segment players evaluated on a single BiLSTM. Both runtimes scale with the number of coalition evaluations required by Shapley value computation.

This work. FO reduces the inference count to a single batch of $T \times V + 1 = 301$ windows per instance, no coalition sampling, no permutations, deterministic result. Under the same domain (S&P 500, $T=20$, CPU), FO achieves **17.6 ms/sample** (LSTM), a reduction of $\times 159.8$ over TimeSHAP and $\times 67.8$ over GS-SHAP. Table 3 summarises all three concrete advances over prior work with verifiable numbers.

Table 3. Concrete advances over prior work. †: reported by Kim and Park [9], Table C3 (S&P 500, $T=20$, $M=50$, CPU). ‡: measured in this work, same domain and T . *: different synthetic setup ($T=50$, $V=4$) vs. ours ($T=20$, $V=15$); same function class.

Prior work	Open gap	Their result	This work
DynaMask [6]	Requires gradients; no RF support	AUP 0.97*, no RF	AUP 0.9960 , RF ✓
TimeSHAP [4]	Sequential models only; slow	2,813 ms/sample†	17.6 ms/sample ‡ ($\times 159.8$)
GS-SHAP [9]	Grouped output ($K \times J$); single model	1,194 ms/sample†; $K \times J$ only	17.6 ms/sample ‡; $T \times V$ ✓

5.4 Model-Agnosticism Verification

A key claim of this work is that the framework generates *model-specific* explanations — attributions that reflect the internal behavior of the queried model, not an artefact of the perturbation method. This is evaluated through two complementary tests on 100 randomly selected test instances.

Inter-model consistency (Spearman ρ). For each pair of models, we compute the mean Spearman ρ between their flattened $\hat{a}$ vectors over all instances (Table 4).

Table 4. Inter-model consistency (Spearman ρ , mean over 100 instances). High within-family, lower across-family correlation confirms that attributions capture model-specific behavior.

Model pair	Spearman ρ
LSTM – Transformer	0.748
LSTM – Random Forest	0.255
Transformer – Random Forest	0.382

The LSTM–Transformer pair yields $\rho=0.748$: both neural architectures agree on the most salient importance patterns despite differing mechanisms. Neural–

RF pairs yield $\rho < 0.40$, reflecting qualitatively different information processing by the ensemble. This cross-family divergence is only observable under a framework that runs on all three architectures simultaneously; DynaMask requires gradients (precluding RF) and TimeSHAP requires sequential structure (precluding RF).

Statistical significance (Mann-Whitney U). Attributions are compared against L_1 -normalised uniform random matrices of identical shape using the Mann-Whitney U test. All three models reject the null hypothesis of equivalence with the random baseline by tens of orders of magnitude ($U=3.55 \times 10^9$ LSTM, 3.59×10^9 Transformer, 2.72×10^8 Random Forest; $p < 10^{-300}$ in all three cases), confirming that the framework produces informative, non-trivial attributions regardless of model nature.

5.5 Robustness Checks

The tests above use a single reference vector, a single target formulation, and a random test sample. This subsection reports three further checks reusing the checkpoints already described, plus a log-return retraining.

Naturally bounded target. Retraining all three models on next-day log return instead of `Close` (same split, scaler, architectures) removes the MinMax saturation flagged in Section 5.2. Under this bounded target, *no* model beats a zero-return persistence baseline (MSE= 0.00257; LSTM 0.00268, Transformer 0.00260, RF 0.00446), consistent with weak-form market efficiency. The RF-vs-neural MSE gap shrinks from $\sim 35\times$ to $\sim 1.7\times$, and inter-model consistency reproduces the same pattern (LSTM–Transformer $\rho=0.541$, LSTM–RF $\rho=0.187$, Transformer–RF $\rho=0.311$; $p \approx 0$ vs. random), confirming agnosticism verification is not an artefact of price autocorrelation.

Directional accuracy. Direction-of-movement accuracy against the previous close, for both targets, falls within a narrow band around chance for all six model-target combinations (Close: LSTM 47.1%, Transformer 46.4%, RF 48.0%; Log_Return: LSTM 46.6%, Transformer 50.3%, RF 47.5%), reinforcing the market-efficiency reading above.

Reference sensitivity and market regime. Table 5 reports inter-model consistency under two alternative references (median, zero) on the same 100 instances, and under the two market regimes present in the test period (2022 bear, 188 windows; 2023–2024 bull, 501 windows; the test period excludes the 2020 COVID crash, which falls in the development split).

The neural-vs-ensemble gap holds under every reference and every regime, confirming that the framework’s central finding does not depend on the arbitrary choice of reference vector nor on the market window sampled. Consistency is systematically lower under the bear regime for all three model pairs, suggesting the three architectures agree less on what drives predictions when the market is more volatile. LSTM and Transformer attributions are highly stable across reference choices (mean-vs-median $\rho \approx 0.99$); Random Forest is markedly more sensitive (mean-vs-median $\rho=0.81$, mean-vs-zero $\rho=0.67$).

Table 5. Inter-model consistency (Spearman ρ) across reference vectors and market regimes. The neural–neural pair exceeds both neural–ensemble pairs in every row.

Condition	LSTM–Transf.	LSTM–RF	Transf.–RF
Mean reference (baseline)	0.748	0.255	0.382
Median reference	0.750	0.332	0.475
Zero reference	0.587	0.255	0.370
Bear market 2022	0.703	0.173	0.287
Bull market 2023–2024	0.764	0.278	0.404

5.6 Same-Conditions Quality Benchmark: FO vs. SVS

To quantify attribution quality in the absence of ground truth, we apply the faithfulness protocol of [6]: for each fraction $\alpha \in \{0.1, \dots, 0.6\}$, the $\lfloor \alpha \cdot T \cdot V \rfloor$ highest-attribution cells are replaced by their reference value and the mean absolute prediction shift is measured:

$$\Delta_\alpha = \frac{1}{N} \sum_{i=1}^N \left| f(\mathbf{X}_i) - f(\tilde{\mathbf{X}}_i^{(\alpha)}) \right| \quad (5)$$

A faithful explanation produces high Δ_α : perturbing truly relevant cells significantly affects model predictions.

SVS (Shapley Value Sampling, 15 permutations) is the only other method that is model-agnostic, produces a $T \times V$ matrix, and runs on all three models, making it the valid same-conditions comparator. Evaluation uses the same S&P 500 test set ($N=100$; SVS over $N=10$ due to computational cost), same reference vector, same three models.

Table 6. Prediction shift $\Delta_{0.3}$ at $\alpha=0.3$. IG not applicable to Random Forest ($\dagger$); SVS: $N=10$, 15 permutations. Bold: best per model.

Model	FO	FP	IG	SVS
LSTM	0.5033	0.4776	0.5034	0.4891
Transformer	0.4518	0.4320	0.4565	0.4352
Random Forest	0.3047	0.2723	$\dagger$	0.3212
FO–SVS: LSTM +0.0142 (FO better) Transformer +0.0166 (FO better) RF –0.0165 (SVS better)				

On neural models, FO surpasses SVS in faithfulness ($\Delta_{0.3}$: +0.014 LSTM, +0.017 Transformer) while running 22–30 $\times$ faster. On Random Forest, SVS outperforms FO by 0.017: coalition sampling captures combinatorial interactions among decision-tree splits that single-cell occlusion misses. The structural source of FO’s speed advantage is that it issues exactly one batched inference call of size $T \times V + 1 = 301$, whereas SVS issues $2 \times n_{\text{perms}} \times T \times V = 9,000$ evaluations at

Table 7. Per-sample runtime (ms, CPU, mean $\pm$ std over 5 runs of 50 samples). FO: $T \times V + 1 = 301$ forward passes. SVS: $2 \times 15 \times T \times V = 9,000$ evaluations.

Model	FO (ms)	SVS (ms)	FO speedup
LSTM	17.6 ± 0.9	469.1 ± 3.1	$\times 26.6$
Transformer	13.8 ± 0.3	413.5 ± 15.9	$\times 30.0$
Random Forest	112.0 ± 3.9	2519.4 ± 82.4	$\times 22.5$

15 permutations. IG surpasses FO on Transformer by 0.005 at the cost of gradient access (precluding RF). FP consistently ranks last: replacing a cell with another observed value introduces exploitable market signal, artificially attenuating the measured perturbation impact.

6 Conclusions

This paper presented a model-agnostic temporal explainability framework for financial time series prediction. Its core contribution is a two-level adapter pattern that decouples the temporal perturbation engine from any specific model backend, enabling any model that exposes a prediction function with signature $(N, T, V) \rightarrow (N,)$ to be connected without modifying any framework component.

Under identical conditions, same test set, reference vector, and three models, FO matches SVS, the only other model-agnostic method producing a $T \times V$ matrix on all three models, in synthetic ground-truth precision (AUP = 0.9960 both) while running 22–30 $\times$ faster on neural architectures and slightly behind on Random Forest (−0.017 faithfulness), where coalition sampling better captures combinatorial tree interactions.

The framework also produces statistically non-trivial attributions ($p < 10^{-300}$) and inter-model consistency coherent with architectural families ($\rho = 0.748$ between neural models vs. $\rho < 0.40$ against Random Forest), a pattern that holds under alternative reference vectors, under both market regimes present in the test period, and under a naturally bounded target that removes the MinMax saturation affecting Random Forest under **Close** (Section 5.5). Neither target formulation beats a naive persistence baseline, in magnitude or in direction, consistent with weak-form market efficiency. The main remaining limitation is evaluation on a single financial index. Future work includes multi-index evaluation and a deeper analysis of why inter-model consistency decreases under volatile market conditions.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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